

OMMC 2026 Main Round

OMMC Staff

September 2026



HOW TO SUBMIT

1. During the testing period, go to our official test portal: <https://ommc-test-portal-new.vercel.app/>. **One person submits per team.** Teams can have 1 to 4 people, inclusive.
2. Register an account using Google or Discord.
3. Fill out team registration information and answers.
4. Submit your answers and all answers and information will be saved to your account. **The submission and your registration information are to be sent in TOGETHER.**
5. You can access this information, and **send in an updated submission if you choose,** if you sign back in to your account later within the testing period. **We will take the last set of answers submitted.**

NO, you do not need to “sign up” beforehand in the conventional sense. Just sign into the test portal **ANYTIME DURING THE TESTING PERIOD** and **SEND IN A RESPONSE.**

RULES

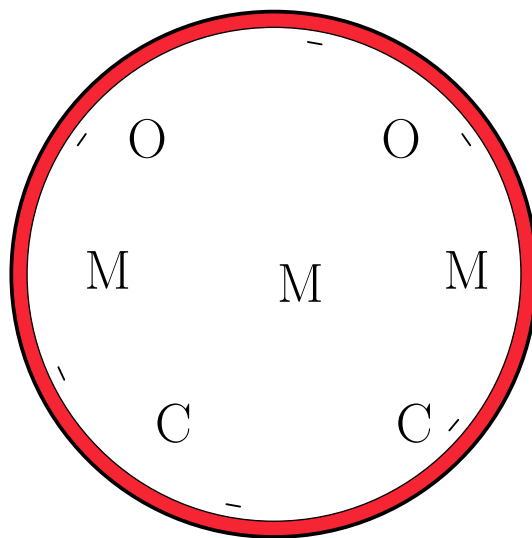
1. This is a **25-question, untimed examination** that can be worked on by a team of up to 4 people. Only one person from each team has to submit answers. If we receive multiple submissions from a team, the most recent submission will be graded.
2. All answers are positive rational numbers inputted as " a/b " for relatively prime positive integers a, b where $b > 1$, or nonnegative integers. Your score will be the number of correct answers; there is neither partial credit nor a penalty for wrong answers. Top scoring teams will move onto the final round.
3. No aids other than writing utensils, scratch paper, rulers, compasses, erasers, and a **four-function calculator** are allowed. The use of **graphing calculators, smartphones, smartwatches, and/or outside websites is NOT ALLOWED**. In particular, **Desmos, Geogebra, Wolfram Alpha, and other similar websites are forbidden**. Failure to follow this rule will result in your test score being voided.
4. Discussion of any aspect of the test outside your team is **not allowed** until submissions are closed and discussion is opened. Failure to follow this rule will result in your test score being voided.
5. Submissions open on **September 12th, 2026, 12:00 PM EDT** and close at **September 19th, 2026, 12:00 PM EDT**.
6. Diagrams are not necessarily to scale.
 - If you do not submit scrap paper, you will **NOT** be considered for the final round. You may still submit without scrap paper. If you choose to submit scrap paper, it **must** be in the following format. ONE person from each team will send ONE email to ommcofficial@gmail.com. The subject line should be "[SCRAP PAPER] (Team Name)".
 - The scrap paper may be submitted in any reasonable format (e.g. do not submit 100 separate screenshots, or walls of text).
 - At OMMC's discretion, we will *selectively interview* some teams before the final round live to determine if they will advance. These interviews will happen on the weekend of Sep 19–20, and will be about the problems.

Any questions on the above should be emailed to ommcofficial@gmail.com. With that, good luck! The OMMC team has spent a lot of time on this contest and we hope that you enjoy your experience.

PROBLEMS

1. Using only the numbers 6 and 7, along with the basic arithmetic operations (addition, subtraction, multiplication, and division) and parentheses, Orang can form the number 67. What is the minimum number of numbers he must use?

2. Find the minimum number of straight-line cuts in the cake below such that each slice contains at most one letter.



3. Let

$$N = \sum_{k=0}^{1434} 2^k \cdot 100^{1434-k} = 100^{1434} + 2 \cdot 100^{1433} + \dots + 2^{1434}.$$

Find the 67th digit from the left in the decimal representation of $14N$.

4. Find the number of sequences of real numbers $x_1, x_2, \dots, x_{30}$ with $x_{30} = x_1$ satisfying

$$x_i^3 + x_{i+1}^3 + 1 = 3x_i x_{i+1}$$

for every $i = 1, 2, \dots, 29$.

5. Let $\text{rev}(n, k)$ denote the reversal of the k -digit binary representation of an integer n , including leading zeros. For example, $\text{rev}(11, 6) = 110100_2$ since $11 = 001011_2$. Find the number of integers n with $0 \leq n < 2^{12}$ such that

$$n + \text{rev}(n, 12) < 2^{12}.$$

6. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be defined by $f(z) = az$, where $a \in \mathbb{C}$ is fixed. Suppose that for every complex number z ,

$$|f(z)| = |f(z) - \sqrt{2026} \cdot z| = \left| \sqrt{2026} \cdot (iz) + f(z) \right|.$$

Find $|f(a)|$.

7. Let $ABCD$ be a convex cyclic quadrilateral such that the areas of $\triangle ABC$, $\triangle BCD$, and $\triangle CDA$ are in the ratio $3 : 5 : 4$. Determine the minimum possible value of $x + y + z$ if x, y, z are positive integers such that

$$x \cdot AB^2 + y \cdot BC^2 = z \cdot AC^2$$

holds for every such quadrilateral $ABCD$.

8. Let P be the set of all permutations of $\{1, 2, 3, 4, 5\}$. For any permutation $\pi \in P$, define $S(\pi)$ to be the set of all distinct permutations that can be generated by repeatedly composing π with itself. That is,

$$S(\pi) = \{\pi, \pi^2, \pi^3, \dots\}.$$

As π ranges over all permutations in P , how many distinct sets $S(\pi)$ are formed?

9. Determine the number of monic polynomials $P \in \mathbb{Z}[x]$ of positive degree such that, for every positive integer n ,

$$P(n)^2 + 2(3n^2 + 3n + 1)P(n + 1) - (3n^2 + 3n + 1)^2$$

is the square of an integer, and such that

$$|P(1) + P(4) + P(34)| < 209.$$

10. Let a, b, c, d be complex numbers such that

$$a + b + c + d = 4, \quad ab + cd = 5, \quad ac + bd = 6, \quad ad + bc = 7.$$

Find the product of all distinct possible values of

$$a^3 + b^3 + c^3 + d^3.$$

11. Let n be a positive integer. Eric chooses uniformly at random from the set

$$\left\{0, \frac{1}{10^n}, \frac{2}{10^n}, \dots, \frac{10^n - 1}{10^n}\right\}.$$

For every positive integer i , he writes down the result of rounding x to i decimal places. For example, when $n = 4$ and $x = 0.6741$, Eric writes down 0.7, 0.67, 0.674, and 0.6741 for every $i \geq 4$. Notice that among the numbers Eric writes down, both digits 6 and 7 occur.

Let a_n be the probability that Eric writes down a number containing at least one of the digits 6 or 7 at some point. Given that

$$\lim_{n \rightarrow \infty} \sqrt[n]{1 - a_n} = \frac{a + \sqrt{b}}{c}$$

for positive integers a, b, c with $\gcd(a, c) = 1$, compute $a + b + c$.

12. Let ABC be a triangle such that $AB = 13$, $BC = 14$, and $AC = 15$. The internal angle bisector of $\angle ABC$ intersects the altitude from A and the perpendicular bisector of BC at D and E , respectively. Let $F = AE \cap BC$ and $G = DF \cap AB$. Find the area of hexagon $AECBGD$.
13. Allison generates a sequence of integers $a_1, a_2, \dots, a_{179}$, where each term a_k is chosen independently and uniformly at random from the set $\{1, 2, \dots, k\}$ by her cat Snowball. After generating the sequence, Allison looks at each pair of integers $1 \leq x < y \leq 179$. She writes down the pair (x, y) if both x and y occur an odd number of times among $a_1, a_2, \dots, a_{179}$. Find the expected number of pairs Allison writes down.

14. Let S be a set of n lattice points in the Cartesian plane such that S contains 7 points forming the vertices of a strictly convex heptagon, and for any 5 points in S that form the vertices of a strictly convex pentagon, there is at least one point of S strictly in the interior of this pentagon. Find the minimum possible value of n .

15. Let $ABCDEF$ be a convex cyclic hexagon such that $AB = BC$, $DE = EF$, $AF = 5$, $CD = 7$, $AD = 9$, and $CF = 10$. Find the value of BE^2 .

16. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Suppose that the function $f: \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ satisfies

$$f(nf(m)) = m^2 f(n)$$

for all positive integers m and n . Given that $f(6) = 18$ and $f(10) = 63$, determine the value of $f(300)$.

17. Find the number of ordered pairs of positive integers (x, y) satisfying

$$x^2 + 14y^2 = 6^{2026}.$$

18. Larry the Viking is learning about remainders at school. His teachers mentioned a rule stating that the remainder when a number is divided by a positive integer is the same as the remainder obtained by adding its decimal digits. He forgot which divisor this rule was for, so he tries using the number 2025. Let p be the probability that, for a uniformly chosen positive integer less than 10^{100} , the remainder obtained by adding its decimal digits is equal to its remainder modulo 2025. What integer is closest to $\frac{1}{p}$?

For example, if Larry applies the rule to 2024, then he obtains a remainder of $2 + 0 + 2 + 4 = 8$ modulo 2025. This would be incorrect, since the remainder when 2024 is divided by 2025 is 2024.

19. Let $ABCD$ be a cyclic quadrilateral with circumcenter O , and let M be the midpoint of $\overline{AB}$. Suppose that $\angle CMD = \angle BAD + \angle ABC = 90^\circ$. Given that $AB = 15$ and $CD = 8$, find the area of $ABCD$.

20. Find the sum of all positive integers n for which

$$\frac{((2n-2)!)^2(5^{2n} + 7^{2n} + 11^{2n} + 1729) + 1}{4n-3}$$

is an integer.

21. Consider a 2026×2026 grid where each cell (i, j) , for $1 \leq i, j \leq 2026$, initially contains $i \cdot j$ chips. A move consists of choosing any 2×2 subgrid and removing 1 chip from each of its four cells, provided that each of the four cells has at least one chip. A configuration is called terminal if no further moves can be made; that is, every 2×2 subgrid contains at least one cell with 0 chips. Different valid sequences of moves can lead to different terminal configurations with varying total numbers of moves. Let $M_{\max}$ be the maximum possible number of moves in any valid sequence, starting from the initial configuration and ending in a terminal configuration, and let $M_{\min}$ be the minimum possible number of moves in any such sequence. Find the value of $M_{\max} - M_{\min}$.

22. Let ABC be a triangle with circumcenter $(0, 0)$, incenter $(8, 1)$, and orthocenter $(16, 3)$. If the vertices of ABC are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) ordered lexicographically (first by increasing x -coordinate, then by y -coordinate), compute $x_1y_2 + x_2y_3 + x_3y_1$.

23. Let n be a positive integer. Every second, Culver writes a uniformly random integer from 1 to n on the whiteboard. Let $f(n)$ be the expected number of seconds until the n most recently written integers form a permutation of $1, 2, \dots, n$. Let u, v, w, x, y be unique real constants such that if we define the unique function $g(n)$ that satisfies

$$f(n) = un^v w^n \left(1 + xn^{-1} + yn^{-2} + g(n)\right),$$

then there exists a fixed positive real constant C such that $|g(n)| \leq Cn^{-3}$ for all positive integers n . Evaluate $\lfloor 288(v + x + y) \rfloor$.

Note: Stirling's approximation states that if we define the unique function $h(n)$ satisfying

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \frac{1}{12}n^{-1} + \frac{1}{288}n^{-2} + h(n)\right),$$

then there exists a fixed positive real constant C such that $|h(n)| \leq Cn^{-3}$ for all positive integers n .

24. Alice is on vacation in Atlantic City and hopes to strike it rich by playing double-or-nothing slot machines. Luckily for her, one of her friends, Bob, recently landed a job at the casino, and successfully rewired one of their slot machines to give Alice an advantage when no one was looking. Usually, the slot machines favor the house by doubling the player's money with $\frac{1}{3}$ probability and giving nothing with $\frac{2}{3}$ probability, but the hacked slot machine offers the exact opposite odds. Unfortunately, Bob forgot to tell Alice which machine he rigged out of the two, and, even worse, Alice only has one dollar left to spend: the slot machines' exact minimum bet!

Alice has settled on the following betting strategy: First, she will bet her dollar at one of the machines at random, and then, for each subsequent round, given that she still has money left, she will switch her chosen machine whenever she loses, and stay on it whenever she wins, always betting one dollar at a time. She wants to eventually walk out of the casino with at least a hundred dollars. If p is the probability that she is able to do so, find the integer closest to $100p$.

25. Consider real numbers $a_1, a_2, \dots, a_{2025}$, not all zero, with a sum of 0. Let x and y be the minimum and maximum possible values of

$$\frac{\sum_{n=1}^{2025} a_n^2}{\sqrt{\sum_{n=1}^{2025} a_n^4}}.$$

Evaluate $\lfloor 10^4(x^2 + y^2) \rfloor$.